

Trial-level RNN: how the network knows the rule (2 vs 3 categories)



a_{t-1} : category chosen on the previous trial s_{t-1} : previous bar length r_{t-1} : previous outcome s_t : current bar length cue_t : number of categories shown on screen

Same read-out in both models: two criteria c_1, c_2 on the length axis define $P(a_t)$; in 3-cat they sit apart, in 2-cat they collapse onto one boundary (5.85).

Left: the cue is an input, so the criteria move on the switch trial. Right: the cue is withheld, so a hidden variable has to infer the rule from feedback and the criteria drift over trials.